

nothing added to commit but untracked files present (use "git add" to track)
● mnit@mnit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/CryptoLabs\$ python3 secure_application/src/iot_manager.py

```
=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 1
Enter device ID: 232
Enter device name: a
Enter device type: watch
Device registered successfully.
```

```
=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 2
Enter device ID: 232
Device Name: a
Device Type: watch
Status: ONLINE
```

```
=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 6
Goodbye!
```

```

=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 1
Enter device ID: 232
Enter device name: a
Enter device type: watch
Device registered successfully.

=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 4
Enter device ID: 232
Enter configuration: time=12:00
Configuration updated.

=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 6
Goodbye!

• mnit@mnit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/CryptoLab$ echo "TEST FIRMWARE FOR WATCH" > secure_application/outputs/firmware/watch.bin
• mnit@mnit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/CryptoLab$ cat secure_application/outputs/firmware/watch.bin
TEST FIRMWARE FOR WATCH
• mnit@mnit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/CryptoLab$ python3 secure_application/src/iot_manager.py

=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 1
Enter device ID: 232
Enter device name: a
Enter device type: watch
Device registered successfully.

=====
IoT DEVICE MANAGEMENT
=====
1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit
Enter choice: 3
Enter device ID: 232
Enter firmware file path: secure_application/outputs/firmware/watch.bin
Firmware uploaded successfully.
Size: 24 bytes

```

```
=====
IoT DEVICE MANAGEMENT
=====
```

1. Register Device
2. View Device Status
3. Upload Firmware
4. Configure Device
5. Check Device Connection
6. Exit

Enter choice: 5

Enter device address: 12.0.0.1

PING 12.0.0.1 (12.0.0.1) 56(84) bytes of data.

--- 12.0.0.1 ping statistics ---

1 packets transmitted, 0 received, 100% packet loss, time 0ms

```
mnit@mnit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/CryptoLab$ semgrep --version
1.56.0
```

 A new version of Semgrep is available. See <https://semgrep.dev/docs/upgrading>

 Versions prior to 1.76.0 are no longer supported by Semgrep.dev, please upgrade.

```
mnit@mnit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/CryptoLab$ semgrep scan --config p/security-audit secure_application/src/iot_manager.py
METRICS: Using configs from the Registry (like --config=p/ci) reports pseudonymous rule metrics to semgrep.dev.
```

To disable Registry rule metrics, use "--metrics=off".

Using configs only from local files (like --config=xyz.yml) does not enable metrics.

More information: <https://semgrep.dev/docs/metrics>

Scan Status

Scanning 1 file (only git-tracked) with 225 Code rules:

CODE RULES

Language	Rules	Files	Origin	Rules
python	77	1	Community	225
<multilang>	2	1		

SUPPLY CHAIN RULES

 Sign in with `semgrep login` and run
`semgrep ci` to find dependency vulnerabilities and
advanced cross-file findings.

```

mmit@muit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/Cryptolabs$ git status
• mmit@muit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/Cryptolabs$ mkdir -p secure_application/sast
• mmit@muit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/Cryptolabs$ sngrep scan --config secure_application/sast/rules.yml secure_application/src/iot_manager.py

Scan Status
Scanning 1 file (only git-tracked) with 2 Code rules:

CODE RULES
Scanning 1 file with 2 python rules.

SUPPLY CHAIN RULES
No rules to run.

PROGRESS
100% 0:00:00

2 Code Findings

secure_application/src/iot_manager.py
secure_application.sast.path-traversal-user-file-path
Possible path traversal: user-controlled path is passed to open().

41| file = open(path, "r")
| .....
secure_application.sast.command-injection-os-system
Possible command injection: user input is passed to os.system().

69| os.system(command)

Scan Summary
Ran 2 rules on 1 file: 2 findings.

• A new version of Sngrep is available. See https://sngrep.dev/docs/upgrading
• Versions prior to 1.76.0 are no longer supported by Sngrep.dev, please upgrade.
• mmit@muit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/Cryptolabs$ sngrep scan --config secure_application/sast/rules.yml secure_application/src/iot_manager.py > secure_application/sast/sngrep_report.txt

Scan Status
Scanning 1 file tracked by git with 2 Code rules:
Scanning 1 file with 2 python rules.
100% 0:00:00

Scan Summary
Ran 2 rules on 1 file: 2 findings.

A new version of Sngrep is available. See https://sngrep.dev/docs/upgrading

Versions prior to 1.76.0 are no longer supported by Sngrep.dev, please upgrade.
• mmit@muit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/Cryptolabs$ cat secure_application/sast/sngrep_report.txt

```

```

• mmit@muit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/Cryptolabs$ cat secure_application/sast/sngrep_report.txt

2 Code Findings

secure_application/src/iot_manager.py
secure_application.sast.path-traversal-user-file-path
Possible path traversal: user-controlled path is passed to open().

41| file = open(path, "r")
| .....
secure_application.sast.command-injection-os-system
Possible command injection: user input is passed to os.system().

69| os.system(command)

• mmit@muit-HP-Elite-Tower-600-G9-Desktop-PC:~/Documents/Cryptolabs$

```

Sast rules

```

rules:

- id: command-injection-os-system
  message: "Possible command injection: user input is passed to os.system()."
  severity: ERROR
  languages:
  | - python
  pattern: os.system(...)

- id: path-traversal-user-file-path
  message: "Possible path traversal: user-controlled path is passed to open()."
  severity: ERROR
  languages:
  | - python
  pattern: open(...)

```